

Inferential Statistics: Knowledge Score vs. Gender

Independent Two-Sample T-Test

This report examines whether there is a statistically significant difference in clinical knowledge between Male and Female thalassemia carriers.

Descriptive Statistics

Gender	N	Mean (V3 Score)	Std Dev (V3 Score)	Mean (Weighted V3)	Std Dev (Weighted)
Male	85	10.81	7.57	1.638	4.223
Female	115	11.72	8.59	2.216	4.825

Hypothesis Testing (Welch's T-Test)

We use Welch's T-Test (assuming unequal variances) for robust results.

1. Analysis using the mathematically rigorous Weighted V3 Knowledge Score:

- **T-Statistic:** -0.900
- **P-Value:** 0.3694
- **Conclusion:** Not Statistically Significant ($p > 0.05$). There is no significant difference in knowledge between male and female carriers.

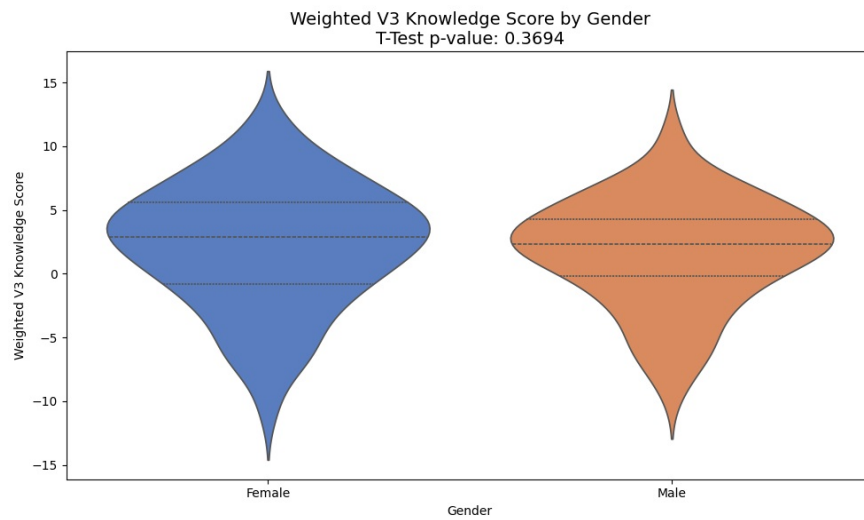
2. Analysis using the standard V3 Knowledge Score:

- **T-Statistic:** -0.793
- **P-Value:** 0.4287
- **Conclusion:** Not Statistically Significant ($p > 0.05$).

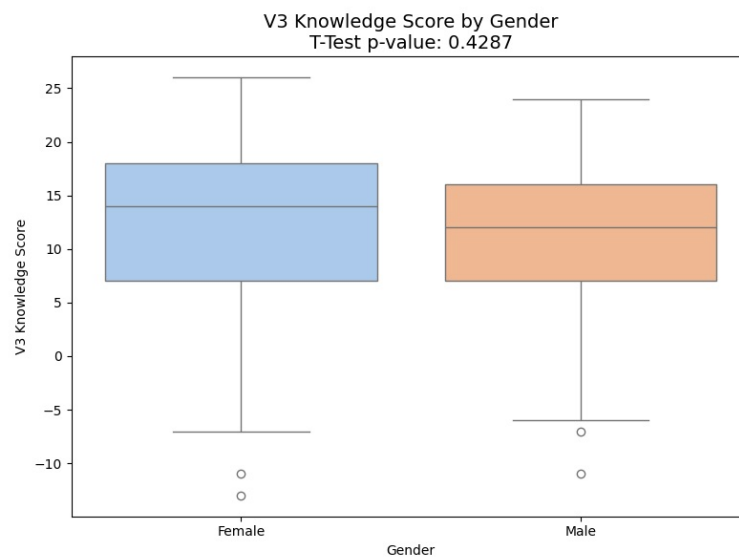
Visualizations

Weighted V3 Score Violin Plot

The violin plot demonstrates the distribution density and quartiles for both genders using the rigorously weighted metric.



Standard V3 Score Box Plot



Box Plot